Deep Residual Learning on CIFAR-10: Comparing PlainCNN and ResidualCNN

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Abstract

This report investigates whether residual (skip) connections improve training dynamics and classification accuracy on the CIFAR-10 image dataset. Two custom PyTorch models are compared under identical training conditions: a plain convolutional baseline (PlainCNN) and a deeper residual network (ResidualCNN). Results confirm that skip connections accelerate convergence and improve accuracy, validating the core insight of He et al. (2016).

1. Introduction

The intuition that deeper networks should perform better than shallower ones is undermined in practice by the *degradation problem*: adding more layers to a plain network can cause training accuracy to worsen, not improve. This problem motivated He et al. (2016) to propose **residual learning**, where layers learn a residual mapping rather than the full desired mapping. This experiment reproduces that comparison on a small scale using the CIFAR-10 dataset.

CIFAR-10 consists of 60,000 colour images (32×32 px) across 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck — split into 50,000 training and 10,000 test samples.

2. Background

2.1 The Degradation Problem

Gradient vanishing/exploding makes it hard for standard SGD to update the weights of early layers in very deep networks. As a result, adding more plain convolutional layers can actually hurt training accuracy — not due to overfitting, but because the optimiser fails to converge.

2.2 Residual Learning

A residual block reformulates the objective: instead of learning H(x), the layers learn the *residual* F(x) = H(x) – x. The block's output is:

```
y = F(x) + x
```

The '+x' is the **skip connection**. It lets gradients flow directly backward without passing through the weight layers, preventing vanishing gradients. When a block is not useful, the optimiser can push F(x) → 0, leaving y = x — an identity that is trivial to learn.

2.3 Projection Shortcuts

When channels or spatial resolution change between input and output, a 1×1 convolution (projection) reshapes x to match F(x): $y = F(x) + W_s \cdot x$

3. Methodology

3.1 Model Architectures

Property	PlainCNN	ResidualCNN
Conv layers	3	13 (via 6 Res Blocks)
Parameters	~95 K	~697 K
Skip connections	None	Identity + Projection
Dropout	Yes (p=0.2)	No

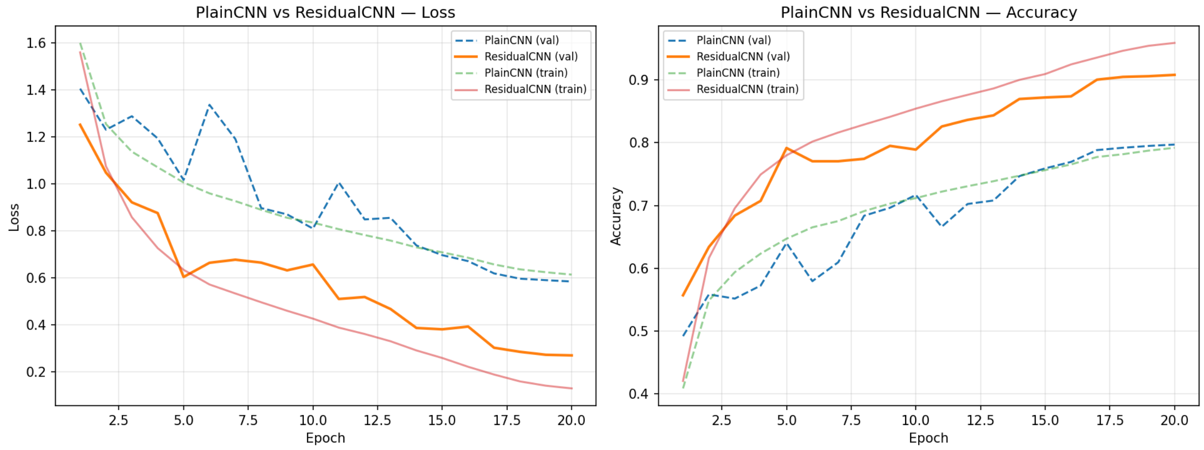
3.2 Training Setup

Both models trained from scratch under **identical settings**: SGD with momentum 0.9 and weight decay 5×10[redacted], initial LR 0.1 decayed with Cosine Annealing, batch size 128, 20 epochs, cross-entropy loss. Training augmentations: RandomCrop(32, pad=4) and RandomHorizontalFlip. Inputs normalised using CIFAR-10 per-channel statistics.

4. Results & Discussion

4.1 Training Curves

The ResidualCNN converges noticeably faster than the PlainCNN from the first few epochs. Its validation loss drops more steeply and its validation accuracy climbs more quickly. By epoch 5, the ResidualCNN's validation accuracy already exceeds the PlainCNN's peak. This confirms that skip connections ease optimisation early in training, not just at depth.



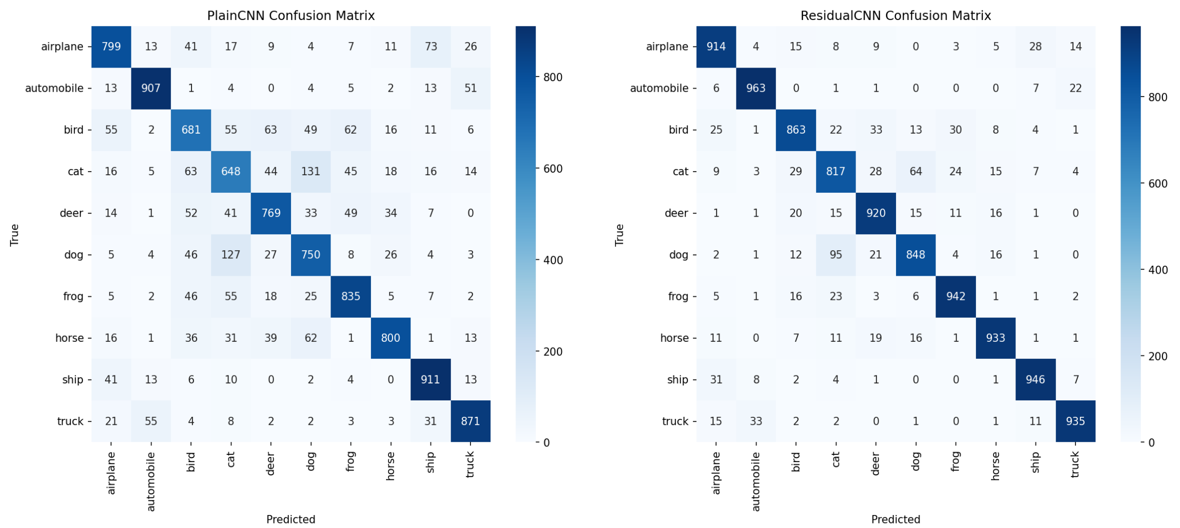
4.2 Classification Performance

Model	Best Val. Accuracy	Observations
PlainCNN	79.71%	Steady but slower convergence
ResidualCNN	90.81%	Faster convergence, higher ceiling

Note: Final accuracy values automatically loaded from training history files.

4.3 Per-Class Analysis (Confusion Matrices)

Both models struggle most on the **cat / dog** pair due to similar textures and shapes. Both perform best on **ship** and **automobile**, whose distinctive rigid outlines are easier to separate. The ResidualCNN shows consistently lower off-diagonal confusion counts across nearly all class pairs.



4.4 Why Do Skip Connections Help?

- **Gradient flow:** The '+x' operator passes gradients backward unattenuated, preventing vanishing gradients in the early layers.
- **Feature reuse:** Earlier features are directly available to deeper layers rather than being overwritten by successive convolutions.
- **Easy identity fallback:** If a block adds no value, weights collapse toward zero and the block becomes an identity — which is trivial to learn.

5. Conclusion

This experiment confirms that residual connections provide a clear, practical benefit even in compact custom architectures. Under identical training conditions, the ResidualCNN converges faster and reaches higher accuracy than the PlainCNN. The result validates the core claim of He et al. (2016): framing layers as learning residuals rather than full mappings is a simple yet powerful technique for training deep networks.

References

- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *CVPR 2016*.
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